

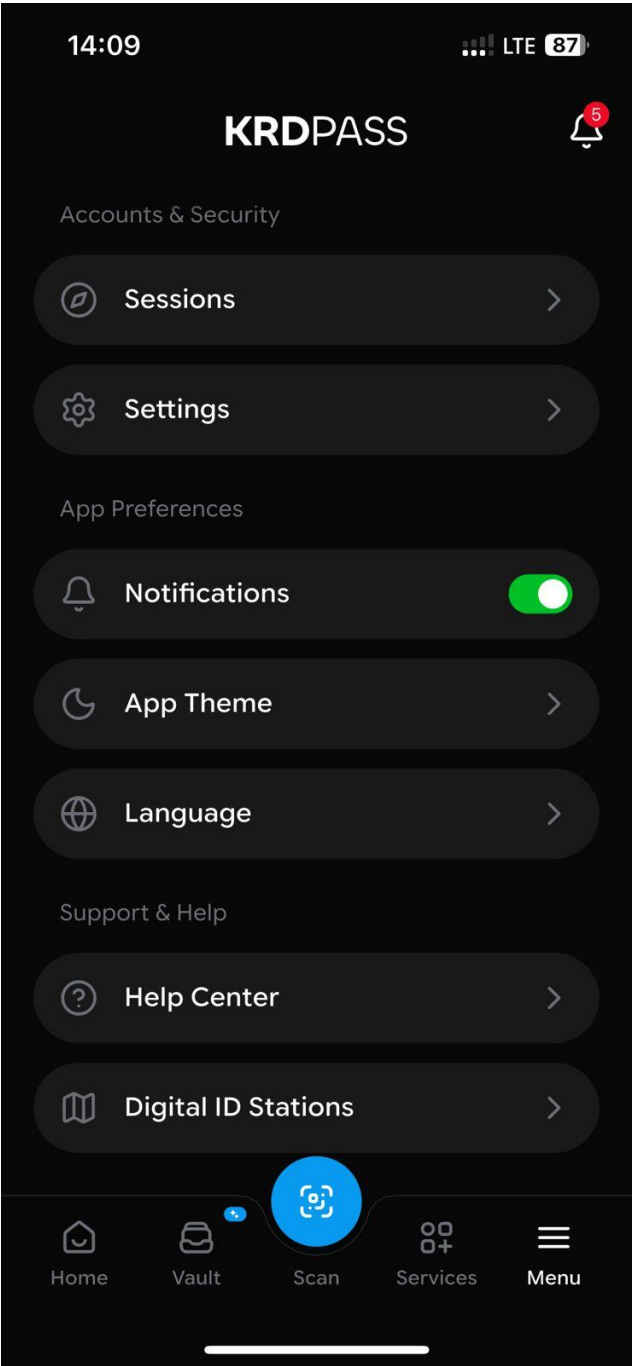
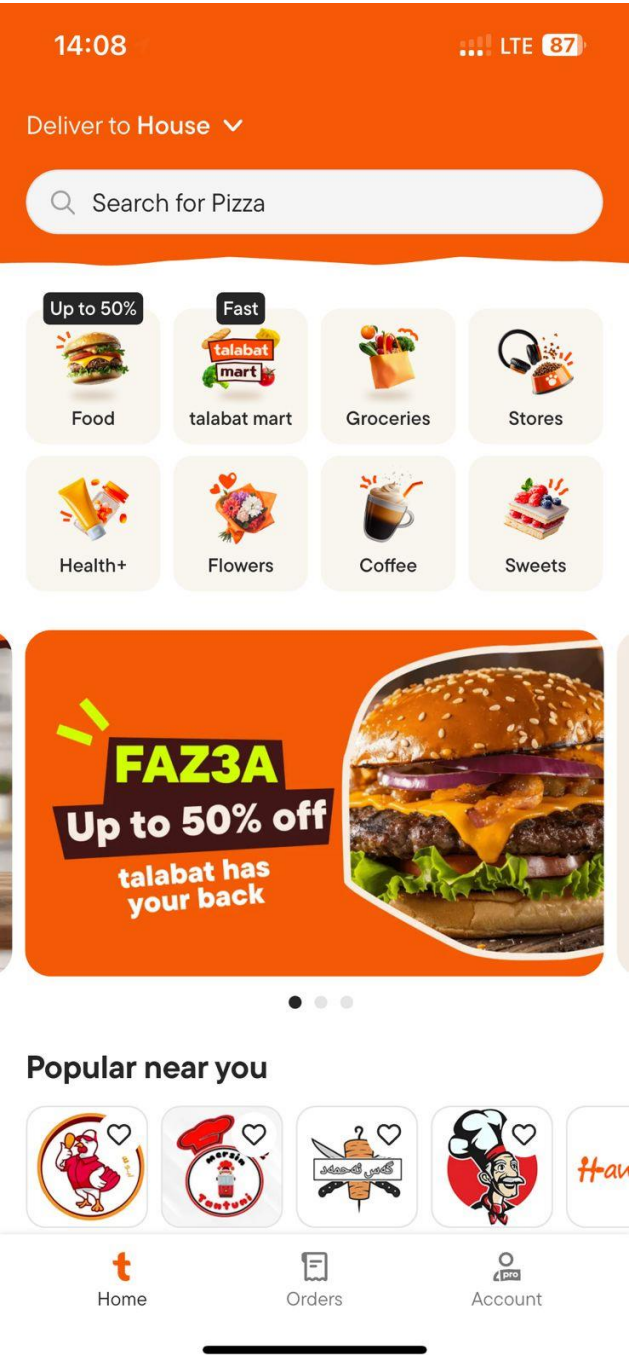
LECTURE 04

Layout Widgets

Row, Column, and Stack — arranging widgets on screen

Instructor · Muhammad Sabah

Layout



Today's plan

01

Row

Arrange widgets horizontally

02

Column

Arrange widgets vertically

03

Stack

Layer widgets on top of each other

04

Practice

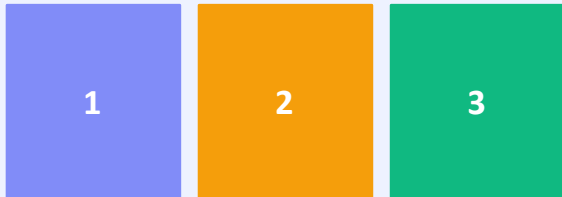
Two hands-on exercises in class

Three ways to lay widgets out

R O W

Side by side

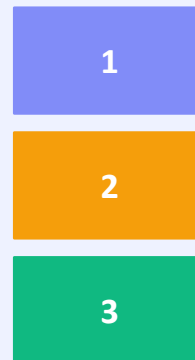
Horizontal direction →



C O L U M N

Top to bottom

Vertical direction ↓



S T A C K

On top

Layered, overlapping



Row — side by side

A Row puts its children in a horizontal line.

CODE

```
Row(  
  children: [  
    Icon(Icons.star, size: 40),  
    Icon(Icons.favorite, size: 40),  
    Icon(Icons.thumb_up, size: 40),  
  ],  
)
```

RESULT ON SCREEN

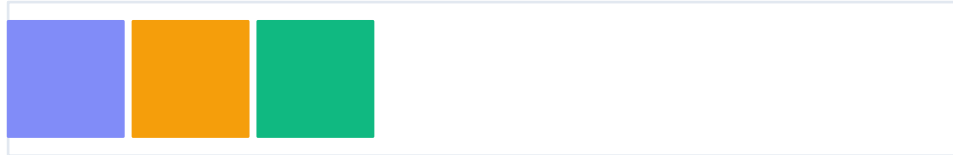


Aligning items in a Row

`mainAxisAlignment` controls how children are spaced horizontally.

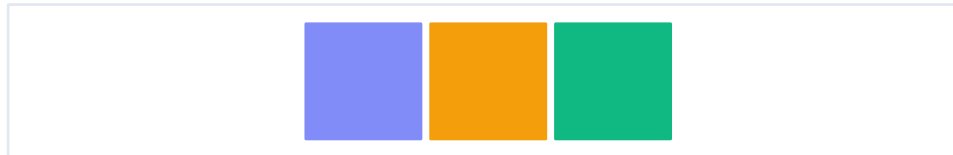
`mainAxisAlignment:`

`.start`



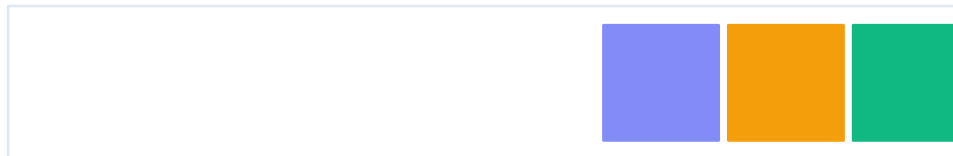
`mainAxisAlignment:`

`.center`



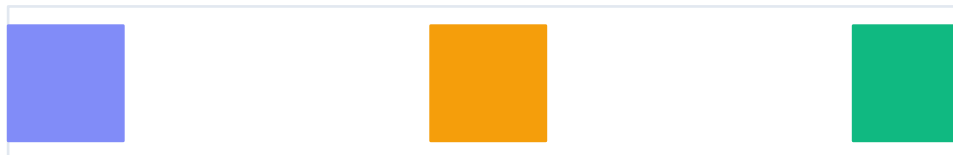
`mainAxisAlignment:`

`.end`



`mainAxisAlignment:`

`.spaceBetween`



Same for `crossAxisAlignment` (vertical alignment inside the Row).

Other values:

- `spaceAround`
- `spaceEvenly`

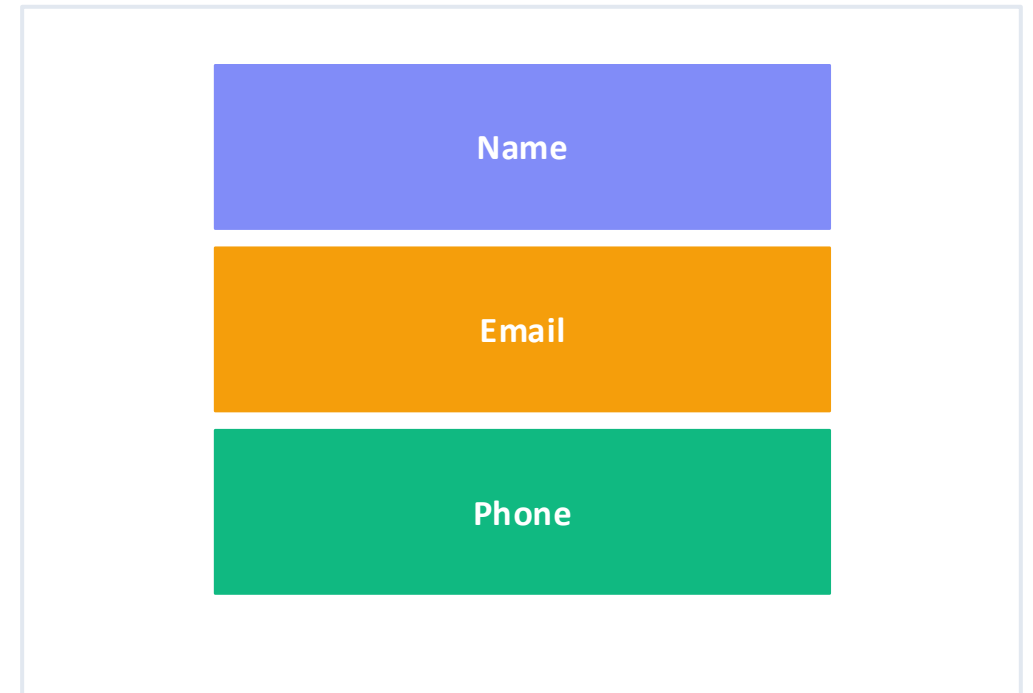
Column — top to bottom

A Column puts its children in a vertical line.

CODE

```
Column(  
  children: [  
    Text('Name'),  
    Text('Email'),  
    Text('Phone'),  
  ],  
)
```

RESULT ON SCREEN

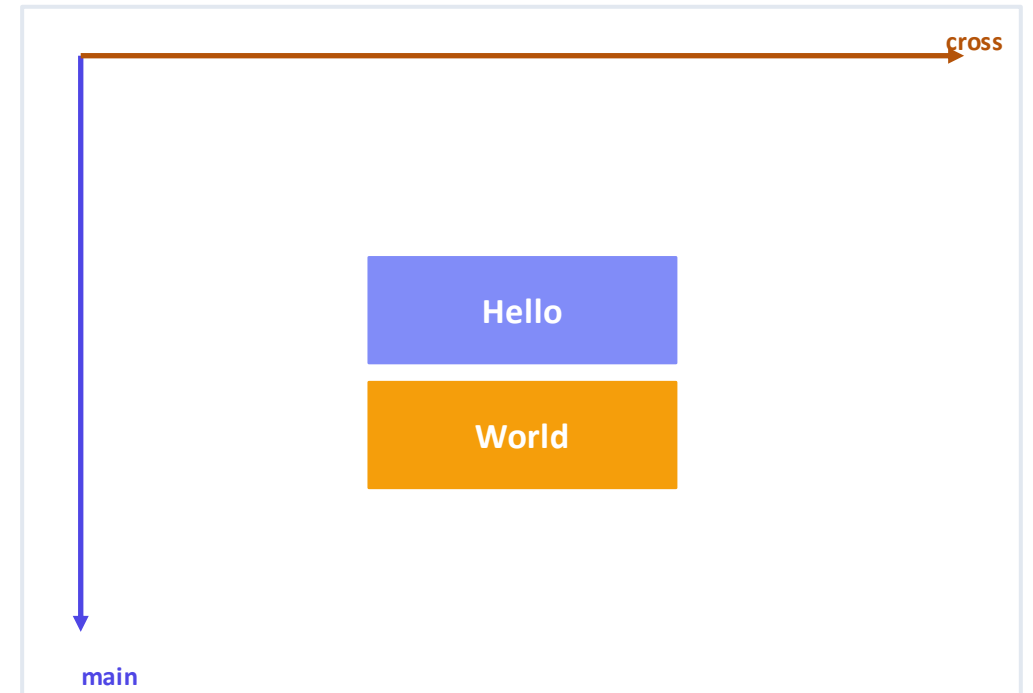


Aligning items in a Column

Same idea — but main axis is now vertical, cross axis is horizontal.

```
Column(  
  mainAxisAlignment:  
    MainAxisAlignment.center,  
  crossAxisAlignment:  
    CrossAxisAlignment.start,  
  children: [  
    Text('Hello'),  
    Text('World'),  
  ],  
)
```

AXES IN A COLUMN



Build a horizontal icon bar

Use Row to display three icons evenly spaced across the screen.

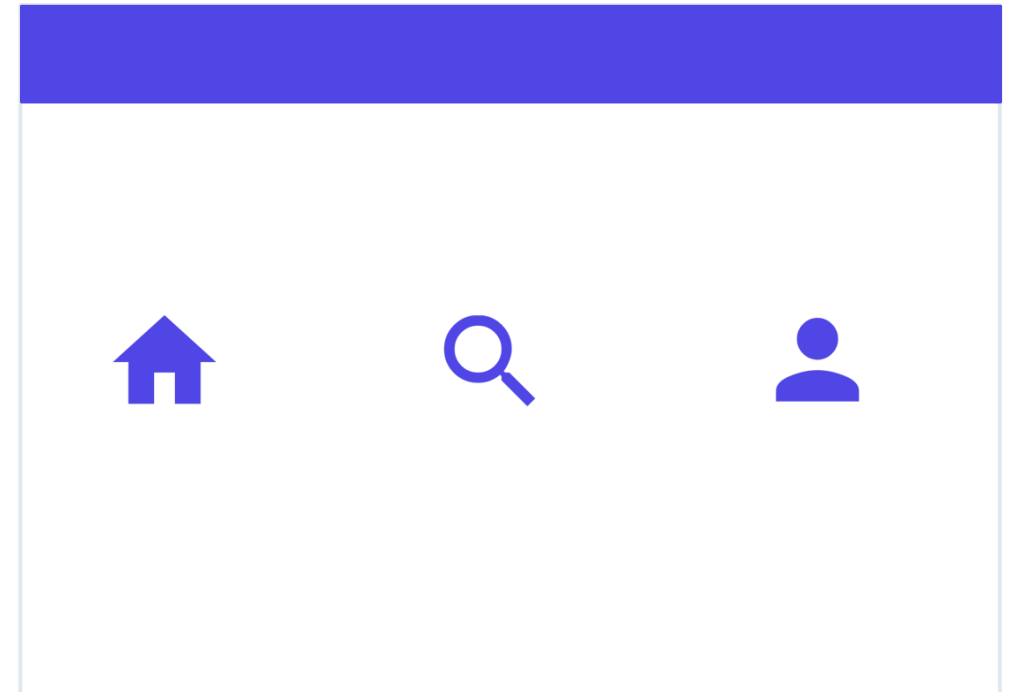
YOUR TASK

1. Create a Row with three Icons
2. Use Icons.home, Icons.search, Icons.person
3. Each icon size 40, color indigo
4. Spread them evenly with spaceEvenly

HINT

```
Row(  
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,  
  children: [ /* your icons here */ ],  
)
```

EXPECTED RESULT



W R A P U P

What we covered today

Y O U N O W K N O W

- ✓ Row — horizontal arrangement
- ✓ Column — vertical arrangement
- ✓ Stack — layering and overlapping
- ✓ mainAxisAlignment & crossAxisAlignment
- ✓ Positioned inside a Stack

T H A N K Y O U

Questions?